

Increased Comprehension of Ranked Choice Voting Leads to Increased Satisfaction with RCV Election Outcomes

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Abstract

Ranked choice voting (RCV) has become a popular reform proposal in the US, but many US voters remain skeptical of its outcomes – especially in elections where the RCV winner differs from the plurality winner. Previous research suggests that voters’ lack of understanding contributes to this skepticism, but existing studies offer little insight into how well voters actually understand the mechanics of RCV, how this understanding can be increased, and whether better understanding would produce greater acceptance of RCV outcomes. We address these issues by conducting a survey experiment with 3,020 U.S. voters, randomly assigning how much information participants receive about the mechanics of RCV and in what format (text, infographic, or video) and then asking participants to assess the RCV and plurality outcome in hypothetical scenarios in which they differ. We find that our treatments (especially the video) significantly improve both objective comprehension of RCV and the perception that the RCV outcome is fairer in these hypothetical scenarios. Our findings suggest that efforts to expand RCV may benefit from clear, engaging explanations of how it works, though further research is needed to understand how best to reach skeptical audiences and promote lasting changes in voter attitudes.

Word count: XXXX

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1 Introduction

Ranked choice voting (RCV) refers to a method of election in which voters rank candidates and the winner is chosen through a process of successively eliminating unpopular candidates and transferring votes according to vote rankings. In recent years, forms of RCV have been adopted to replace plurality or runoff elections for municipal offices in several US cities and federal offices in two US states (Maine and Alaska). Advocates highlight many potential benefits of adopting RCV, including allowing voters to more fully express their preferences, reducing the role of strategic voting, encouraging entry of new candidates, and ensuring the election of candidates with broader support.

Despite these apparent advantages, public support for RCV in the US is not high. Several studies (including [Blais et al. 2021](#), and [Anthony et al. \(2024\)](#)) suggest that voters don't value the opportunity to rank candidates and prefer a simpler plurality vote. Moreover, according to a recent survey experiment ([Cerrone & McClintock 2023](#)), US voters appear not to appreciate one of RCV's most important features, which is that it can elect a candidate who has broad support despite not being the first choice of a plurality of voters. Indeed, [Cerrone & McClintock \(2023\)](#) find that American voters are more satisfied on average with the outcome of a plurality election than with the outcome of a RCV election in comparable circumstances. Notably, [Cerrone & McClintock \(2023\)](#) find this preference for plurality even in the circumstance where the plurality winner lacks the broad support necessary to win under RCV, which is exactly the circumstance that RCV advocates highlight to demonstrate the superiority of RCV.

Important questions remain, however, about *why* so many US voters are skeptical about RCV. As shown by [McCarthy & Santucci \(2021\)](#), support for RCV is clearly associated with partisanship (Democrats being more supportive than Republicans) and age (young people being more supportive). Another important factor, and the focus of our paper, is comprehension of RCV. Several studies show that US voters find RCV rules more challenging to understand than plurality rules ([Donovan et al. 2019](#), [Anthony et al. 2024](#)), and

Cerrone & McClintock (2023) find that voters who report that they are less familiar with RCV also report less satisfaction with it. On the other hand, Anthony et al. (2024) find no effect of experimentally varying the amount of RCV explanation given to survey respondents on their attitudes toward RCV; given that (like all other studies we are aware of) Anthony et al. (2024) do not measure comprehension of RCV, this finding suggests either that longer explanations do not improve understanding of RCV or better understanding of RCV does not produce support. Thus it remains an open question how much of voters' apparent dissatisfaction with RCV is due to lack of understanding versus genuine objections to the system and the outcomes it produces. This leaves a significant gap in both the theoretical and practical understanding of how citizens engage with institutional change.

In this paper, we aim to clarify the relationship between understanding RCV and support for its outcomes. We present findings from a preregistered survey experiment conducted with a quota-based sample of 3,020 U.S. registered voters. After a minimal textual explanation of plurality and RCV, respondents were randomly assigned to receive (1) no additional material, (2) a one-minute video explaining how the RCV winner is determined, or (3) an infographic showing the same concepts in a different way. After receiving these treatments, respondents were asked how well they understood plurality and RCV (self-reported comprehension) followed by a series of factual questions about how each system determines the winner in a set of scenarios (objective comprehension). Finally, participants were asked to view two hypothetical election scenarios in which the plurality and RCV winner would differ (one partisan scenario, one non-partisan scenario) and to indicate whether the RCV outcome or the plurality outcome was more fair, as well as which winner would have a stronger mandate to govern.

Our analysis confirms that US voters are skeptical of RCV on average, with about half of respondents seeing the plurality outcome as fairer in scenarios where it differs from the RCV outcome, but that increasing respondents' understanding of how RCV works substantially raises their appreciation for RCV. Both treatments significantly increase objective comprehension of RCV compared to control (especially the video treatment), though only the

video increases self-reported understanding. Both treatments also increase the extent to which respondents perceive the RCV outcome as more fair than the plurality outcome in our hypothetical scenarios (with a larger effect for the video treatment). OLS analysis shows that higher comprehension of RCV strongly predicts greater support for RCV outcomes (including with covariates). Moreover, under the assumption that our treatments only affect participants’ perceptions of RCV by increasing comprehension, IV estimates show a large effect of comprehension on the extent to which respondents perceive the RCV outcome as more fair or see the RCV winner as having a stronger mandate to govern. Thus our study provides rigorous evidence that better understanding of the mechanics of RCV contributes to greater acceptance of RCV results.

2 Background

Although ranked choice voting (RCV) has been implemented in a growing number of US cities and states, important questions remain about voters’ support for RCV, voters’ ability to understand the system, and how support and understanding are related. Our study breaks new ground in addressing these questions by measuring and experimentally manipulating voters’ understanding of RCV and assessing how understanding affects appreciation for RCV.

One of the central features of RCV highlighted by advocates is that RCV can choose a candidate with broad support in circumstances where plurality might fail to do so. Suppose an election takes place involving three candidates (A1, A2, and B). Suppose that B is the first choice of 40% of voters and the last choice of the remaining 60%, some of whom prefer A1 over A2 and some of whom prefer A2 over A1.¹ Despite being the last choice of a substantial majority of voters (a “Condorcet loser”), B can win a plurality election if support is split evenly enough between A1 and A2 – for example, if 40% vote for B, 35% vote for A1, and 25% vote for A2. By contrast, in an RCV election where 40% rank B first,

¹When the Condorcet loser (here, B) is the plurality winner, this situation is known as a “divided majority” scenario in the literature, e.g. Bouton (2015).

35% rank A1 first, and 25% rank A2 first, A2 would be eliminated and her supporters' votes would be transferred to A1, who would defeat B 60%-40%. To RCV advocates, a scenario like this (a “come-from-behind-victory” in which a candidate with broader support defeats the plurality winner after other candidates are eliminated) highlights the superiority of RCV and the defects of plurality.

Perhaps surprisingly, recent research suggests that many US voters fail to appreciate this feature of RCV. Cerrone and McClintock (2023) ask US voters to evaluate the outcomes of hypothetical elections held under plurality and RCV rules. When asked to evaluate come-from-behind-victory situations like the one described in the previous paragraph, Cerrone and McClintock (2023) find that voters' average satisfaction scores are higher when plurality rules are used and the candidate preferred by a majority loses than when RCV rules are used and the candidate preferred by a majority wins. Cerrone and McClintock (2023) attribute this finding to voters' lack of understanding of, and familiarity with, RCV. As evidence, they show that voters' satisfaction with RCV outcomes is strongly increasing in voters' self-reported familiarity with RCV. Moreover, among voters highly familiar with each rule, the satisfaction gap between RCV and plurality disappears. Consistent with Cerrone & McClintock (2023), Anthony et al. (2021) and Blais et al. (2021) find that US voters asked to vote in hypothetical RCV and plurality elections tend to strongly prefer plurality.

We find it plausible that US voters' apparent dissatisfaction with RCV (particularly in come-from-behind-victory situations) is attributable to lack of understanding, as Cerrone & McClintock (2023) suggest – especially given other research highlighting American voters' difficulties understanding RCV (Donovan et al. 2019, Anthony et al. 2021, Atkeson et al. 2024). But other interpretations of Cerrone & McClintock (2023)'s results are possible, some of which are due to specific features of their design. Respondents in Cerrone & McClintock (2023)'s survey may have given higher scores to plurality partly due to other differences between the plurality and RCV vignettes in the experiment. Notably, the plurality vignettes describe the electoral campaign as “competitive”, while the RCV ones do

not, which could sway the opinion of respondents who value competitive campaigns. Moreover, as the authors note, the RCV vignette makes very clear whether the winner has a plurality of top rankings, while it takes close reading of the plurality vignette (and some political sophistication) to determine whether the plurality winner likely has majority support.² Perhaps more importantly (and as recognized by the authors), the correlation they observe between self-reported familiarity with RCV and satisfaction with RCV outcomes could be explained in part by confounding factors (such as partisanship) that could affect both familiarity and satisfaction with RCV.³

As noted, several papers have provided evidence that US voters struggle to understand RCV elections. [Donovan et al. \(2019\)](#) reports that, compared to voters in municipalities not using RCV, voters in municipalities using RCV report greater difficulty understanding the voting instructions in a recent local election. Election officials in Maine surveyed by [Anthony et al. \(2021\)](#) were skeptical that voters understood RCV rules. [Atkeson et al. \(2024\)](#) report that although 68% of voters in a Santa Fe exit poll find the RCV ballot “not at all confusing”, 16% say it is “very” or “somewhat” confusing, and that these voters were less likely to fully rank the candidates. Like [Cerrone & McClintock \(2023\)](#), all of these studies rely on voters’ assessment of their own understanding (or, in the case of [Anthony et al. \(2021\)](#), others’ assessments of voters’ understanding). Voters’ self-reported understanding is doubtless an important consideration in evaluating the suitability of electoral reforms, but without measures of voters’ actual understanding of RCV we don’t know whether self-reported understanding is a good proxy for actual understanding.

Relatedly, existing work offers little guidance about whether and how voters’ understanding of RCV can be efficiently improved. In a US survey experiment, [Anthony et al.](#)

²This is arguably a good design feature because it reflects information typically available to voters in actual elections. [Cerrone & McClintock \(2023\)](#) find that voters are more satisfied with the RCV outcome when the RCV winner is the leader in first choice votes (i.e. when there is no “come-from-behind”) victory. This preference could make sense for someone who understands and likes RCV but prefers outcomes in which the winner has stronger first-preference support from the electorate: in the “come-from-behind” scenario the winner has 45% first-preference support, compared to 46% otherwise.

³Similarly, [Blais et al. \(2021\)](#) find that older people are less interested in more expressive ballot formats (including a ballot involving ranking), which they interpret as evidence that voters prefer a ballot type more strongly when they are more familiar with it. Other differences between older and younger voters could of course explain the pattern.

(2024) randomly vary whether participants read a short text explaining RCV (or plurality, depending on the treatment arm) or a longer text. They find little difference in resulting satisfaction with RCV across the treatment groups. They do not measure whether the text affects understanding of RCV, however, so we do not know if learning took place or not. We also do not know if other information treatments (such as infographics or video) would be more effective than text passages at building understanding and shifting attitudes.

This study aims to address these gaps by experimentally testing whether materials such as videos or infographics can improve RCV comprehension (measured through comprehension questions rather than only self-reports), and in consequence, satisfaction with RCV outcomes – especially in circumstances where the RCV outcome would likely conflict with the plurality outcome.

Although previous literature makes it seem reasonable that voters would appreciate RCV more if they understood it better, there is reason to think that such an effect might be small. Perceptions of RCV have become highly partisan (e.g. [Anthony et al. 2021](#)). Even if a voter receives our educational materials and has an improved understanding of how RCV works, their evaluation and satisfaction with RCV may still mainly depend on their political affiliation and whether they believe that the system will support or harm their party’s future electoral success. Some voters who don’t have a good understanding of RCV could also become more skeptical about it when they learn more about how it works, e.g. that lower rankings only matter when one’s top choice has been eliminated.

By experimentally varying the mode of instruction and carefully measuring both objective understanding and subjective confidence of RCV, our study will provide clearer evidence on whether targeted educational interventions can improve understanding of RCV and, in turn, increase satisfaction with RCV outcomes. In doing so we hope to contribute to the broader conversation about the drivers of citizens’ views on institutional reforms and how support for such reforms can be built through education initiatives and accumulated exposure.

3 Survey design

3.1 Sample and representativeness

We recruited 3,020 U.S.-resident registered voters through CloudResearch Connect, enforcing quotas on political party (36% Democrat, 33% Republican, 31% Independent/Other) and sex (52% female, 48% male) to approximate the 2022 Cooperative Election Study (CES) margins for US voters.

Although the use of an online convenience panel limits the generalizability of our findings, especially to non-internet users or low-information voters, our random assignment to treatment conditions preserves internal validity. Moreover, the application of demographic quotas enhances representativeness on observable characteristics that are likely to correlate with political attitudes, thus providing a meaningful test of the effects of our information treatments on a politically relevant population.

Our sample is broadly representative of the US electorate, with important exceptions. Our sample is notably more educated than the general US population, with 41% of respondents holding a Bachelor’s degree (compared to around 22% of the population) and also younger (with e.g. about 31% of respondents in their thirties, compared to around 18% of the population, and 3% of respondents over 70, compared to 15% of the population). [TODO: table or figure on representativeness]

Our pre-analysis plan (PAP) stated that our analysis would focus on the unweighted sample and not exclude any respondents. Following our PAP, the Appendix also provides our primary results (effect of treatment on comprehension, effect of treatment on perceived fairness of RCV outcomes) with population weights designed to match CES margins on income, age, party affiliation, sex, race, and education. It also provides the same analysis excluding respondents flagged as possible bots.⁴ Both sets of analysis closely resemble our main results.

⁴Specifically, that analysis excludes respondents with low reCAPTCHA v3 scores (below 0.7, around 4.5%) of the sample. reCAPTCHA v3 is a Google bot detection API that assigns a score based on each website visitor’s pattern of mouse and keyboard inputs.

3.2 Information treatments

After indicating whether they have heard of RCV or voted in an RCV election, our respondents are randomly assigned to one of three treatment groups that vary in the degree and type of RCV explanation the respondent receives. We will refer to these groups as *Control*, *Video*, and *Infographic*.

All respondents first read a minimal description of how plurality and RCV operate. The description reads,

In a **plurality** election, you vote for one candidate. The candidate who wins the most votes wins the election.

In a ranked-choice voting election (**RCV**), you rank candidates to indicate your first choice, your second choice, and so on. If your preferred candidate is eliminated, your vote is transferred to your next choice. This process of eliminating candidates and transferring votes continues until a candidate has a majority of votes and that candidate wins the election.

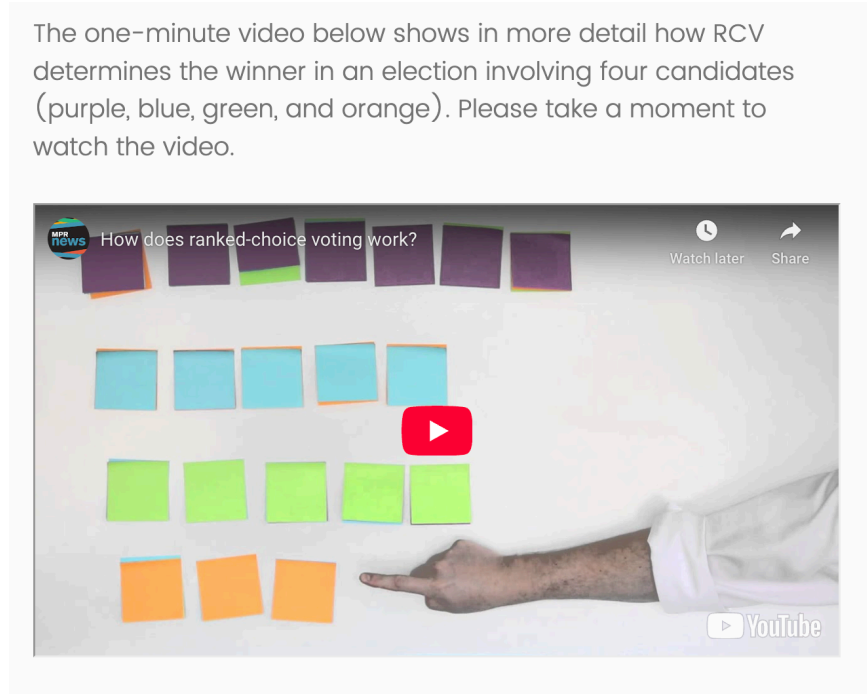
Participants assigned to the *Control* group read only this text. This condition serves as the baseline for evaluating the added effect of richer informational content that expands on and reinforces the points in the control text, explaining how the RCV algorithm works in more depth.

Participants randomly assigned to the *Video* group watch a one-minute video produced by a media organization⁵ that explains the mechanics of RCV in a mock election where the candidates are represented by colors and voters' ballots are represented by stacks of colored sticky notes. (See the screenshot in Figure 1.) The clip highlights how ballots are transferred from eliminated candidates to remaining ones, with voice narration underscoring key points about majority thresholds and vote reallocation rule. The winner of the mock RCV election is a color that was initially in second place (a “come-from-behind” victory).

⁵MPR News, “How does ranked-choice voting work?” <https://www.youtube.com/watch?v=oHRPMJmzBBw>

Crucially, the video contains no editorializing in favor of or against RCV as a reform, suggesting that any effects of the video on perceptions of RCV should operate only through helping viewers understand how RCV works.

Figure 1: Video treatment

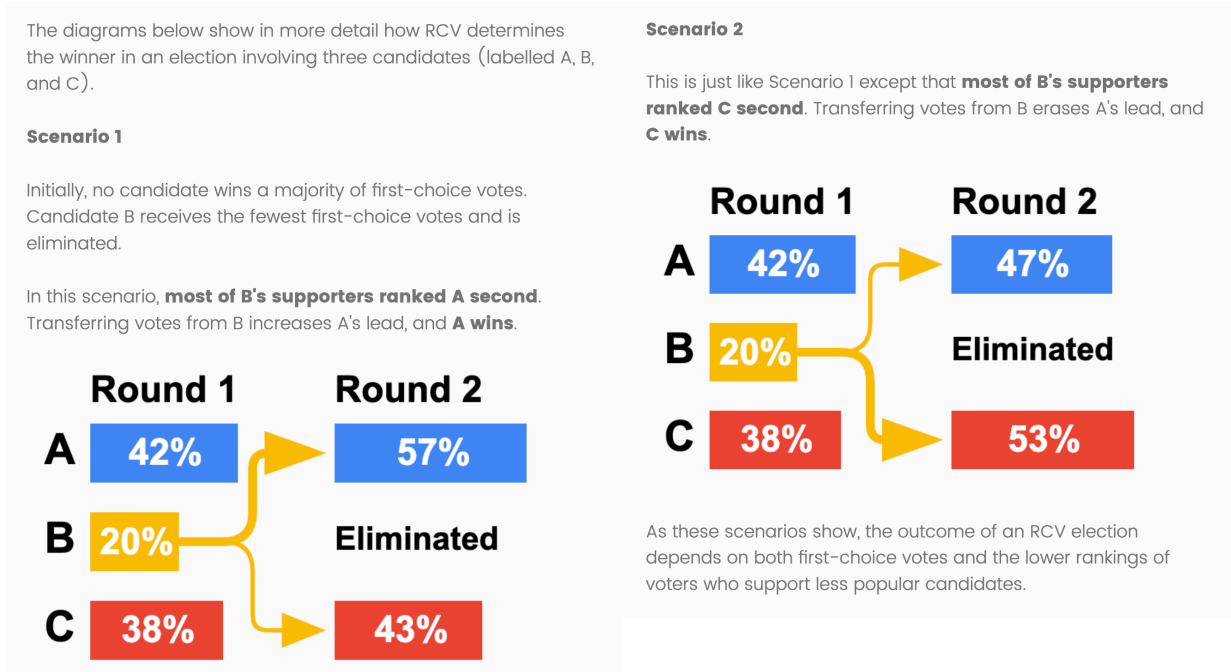


Participants assigned to the *Infographic* group instead see a page of text and infographics using two three-candidate scenarios to illustrate how votes might be transferred in an RCV election. (See Figure 2.) In the first scenario the transfers cause the initial leader to win. In the second scenario the transfers cause the initially trailing candidate to win (another “come-from-behind” victory, CFBV). This treatment highlights how lower-ranked preferences affect results and determine whether a CFBV occurs.

3.3 Measuring comprehension

Post-treatment, we ask respondents to assess their own understanding of RCV (and plurality) on a five-point scale from “Not well at all” to “Perfectly well”. This allows for comparability with previous research based on self-reported understanding of RCV (e.g. [Donovan et al. 2019](#), [Atkeson et al. 2024](#)). We convert this measure to integer steps and

Figure 2: Infographic treatment



rescale it to 0–1.

In addition, we administer six multiple-choice comprehension questions, four on RCV rules and two on plurality voting. We include the plurality questions to discourage respondents from (incorrectly) inferring that we are RCV advocates who hope that they praise RCV later in the survey. The RCV questions assess respondents' understanding of how the winner is determined in an RCV election. Figure 3 shows two of the four RCV questions.⁶

3.4 Measuring satisfaction with RCV vs plurality outcomes

Finally, we ask respondents to assess the outcomes of two hypothetical elections, one partisan (involving a Republican, a Democrat, and either a Green Party candidate or a Libertarian) and one non-partisan (involving five candidates labeled just “A” through “E”). (See Figure 4.) In both cases we provide a table showing percentages for each candidate (but not specifying if these are plurality votes or shares of top rankings). We then stipulate that the candidate shown as leading would win in a plurality election while the second-place

⁶One other question assesses who wins when a candidate initially has a majority of first preferences and the other asks who happens to a voter's ballot when that voter's top-ranked candidate is eliminated.)

Figure 3: RCV comprehension questions. (Correct answers: “Not enough information” and “Candidate B”)

Suppose the results of a **ranked-choice voting (RCV)** election are as follows:

Candidate	A	B	C
% Of First-Choice Votes	40%	35%	25%

Which candidate **wins** the election?

☐ Candidate A

☐ Candidate B

☐ Candidate C

☐ Not enough information

☐ Not sure/prefer not to answer

Suppose the results of a **ranked-choice voting (RCV)** election are as follows:

Candidate	A	B	C	D
% Of First-Choice Votes	40%	35%	20%	5%

All of C's supporters **rank B second**.

Which candidate **wins** the election?

☐ Candidate A

☐ Candidate B

☐ Candidate C

☐ Candidate D

☐ Not enough information

☐ Not sure/prefer not to answer

candidate would win in an RCV election. We ask respondents to assess which outcome would be more fair and which of the two potential winners would have a stronger mandate to govern. In the partisan scenarios, we ensure that the stipulated come-from-behind victory is plausible by having the minor-party candidate be potentially ideologically aligned with the second-place candidate – Green with Democrat, Libertarian with Republican.⁷

We code the responses to these questions as -.5 if the respondent favored the plurality outcome, .5 if the respondent favored the RCV outcome, and 0 if the respondent chose “Not sure/prefer not to answer”. By averaging these measures across the two scenarios, we create a “fairness index” capturing the extent to which the respondent considers the RCV outcome more fair compared to the plurality outcome in a scenario in which the two outcomes differ. We also construct a “mandate index” in the same manner that captures the extent to which the respondent considers the RCV winner to have a strong mandate to govern (compared to the plurality winner) in the same circumstances.

Note that, drawing on [Cerrone & McClintock \(2023\)](#), our outcome measures focus on

⁷We randomize which minor party candidate is shown, but if it is Green, then the Democrat is in second place. We also randomize the vote share of the second- and third-place candidate, using either 44%/9% or 35%/18%, and we randomize response options so that the RCV outcome is equally likely to be listed first for second.

Figure 4: Evaluative judgments. (Not shown: additional question about which candidate has a stronger “mandate to govern”)

Suppose an election takes place in your state.

The results are as follows:

Republican	Democrat	Green
47%	44%	9%

Suppose that

- In an RCV election, the Democrat would win.
- In a plurality election, the Republican would win.

Leaving aside your partisan preferences, which outcome do you think would be more fair?

☐ The Democrat winning (by RCV rules)

☐ The Republican winning (by plurality rules)

☐ Not sure/prefer not to answer

Suppose a nonpartisan election takes place in your state.

The results are as follows:

A	B	C	D	E
37%	32%	18%	10%	3%

Suppose that:

- In an RCV election, Candidate B would win.
- In a plurality election, Candidate A would win.

Which outcome do you think would be more fair?

☐ Candidate A winning (by plurality rules)

☐ Candidate B winning (by RCV rules)

☐ Not sure/prefer not to answer

come-from-behind-victory (CFBV) scenarios in which plurality and RCV would produce different outcomes (assuming similar voting behavior). While [Cerrone & McClintock \(2023\)](#) asks voters to assess either a plurality or a RCV scenario, and infers relative satisfaction by comparing across subjects, we ask each voter to evaluate whether they prefer the RCV outcome or the plurality outcome in a single scenario where they are told that these would be different. Asking the question in this way (compared to asking respondents to evaluate a newspaper-like description of a single election outcome) may reinforce the hypothetical nature of our vignette (arguably, a disadvantage), but it ensures that respondents evaluate the outcome itself and not other extraneous aspects of the vignette (such as the length of the description), and it ensures that all respondents realize that the two rules would produce a different outcome and the plurality winner likely lacks majority support. By eliciting each voter’s preference between the RCV outcome and the plurality outcome, we also gain statistical efficiency compared to [Cerrone & McClintock \(2023\)](#)’s between-subjects design, which requires comparing the mean evaluation of RCV and plurality outcomes estimated in separate subsamples.

4 Results

4.1 Comprehension

Table 1 shows the result of regressing our measure of self-reported RCV comprehension (columns 1 and 2) and our measure of objective RCV comprehension (columns 3 and 4) on our treatment indicators, with and without controls (columns 2 and 4 vs columns 1 and 3).⁸ As expected given randomized treatment and a large sample, the estimated effect of treatment on comprehension barely depends on the inclusion of covariates. Strikingly, the video treatment increases self-reported understanding of RCV by about .062 on a 0-1 scale (or about a third of a step on the original five-point Likert scale) compared to the control group, while the infographic treatment *decreases* self-reported understanding of RCV by about half as much.⁹ Nonetheless, both treatments increase objective understanding of RCV as measured by our comprehension questions. On average, participants in the video group answered 0.4 more questions correctly than those in the control group, while those in the infographic group answered 0.26 more questions correctly; both effects are statistically significant at the 0.001 level and remain robust when we reweight our sample to match the covariate distribution of US voters (see Appendix).

We speculate that the infographic treatment may have decreased confidence while increasing understanding because it exposed gaps in some respondents' prior assumptions or confused respondents about aspects not covered in the comprehension questions we asked. This discrepancy between what participants thought they learned and what they actually learned is consistent with some previous research in cognitive psychology (Koriat 1997, Bjork et al. 2013), and highlights drawbacks of using self-reported understanding as a proxy for actual understanding. (The video, in contrast, may have increased both subjective and objective understanding because it delivers the same basic information in a more gentle, digestible format.) Table 15 in the Appendix shows that the effect of the

⁸Standard error estimation in all regressions is robust to heteroskedasticity (HC2).

⁹With population weights (Table 12), the video effect is insignificant while the infographic effect remains negative and significant.

information treatments on comprehension of RCV is similar for Democrats, Republicans, and others, with slightly larger point estimates for Republicans.

An important question in our analysis is the relationship between the objective and subjective comprehension measures. Across the full sample, we observed a moderate positive correlation between our objective and subjective measures of RCV comprehension (approximately 0.25). This means that individuals who reported greater understanding of RCV tended to perform better on objective measures, but the relationship was far from perfect. Many participants who expressed high confidence answered few comprehension questions correctly, and vice versa. This misalignment underscores the importance of including both types of measures in research on political learning and engagement. Relying solely on self-reported understanding, as some prior studies have done, risks drawing wrong conclusions about how well individuals actually grasp concepts.

The coefficients in columns 2 and 4 of Table 1 highlight some interesting partial correlations between respondent characteristics and understanding of RCV. Older respondents appear to be less confident of their understanding of RCV than younger respondents (conditional on covariates), but they actually perform better on the comprehension questions (again conditional on covariates). Women were substantially less confident but performed only slightly less well (conditional on covariates). Differences in comprehension were small across respondents with different political affiliations (conditional on covariates), which is notable given substantial partisan differences in attitudes toward RCV (see below). Both self-reported and objective understanding are slightly increasing in education, but the differences are larger for the objective measure (conditional on covariates).

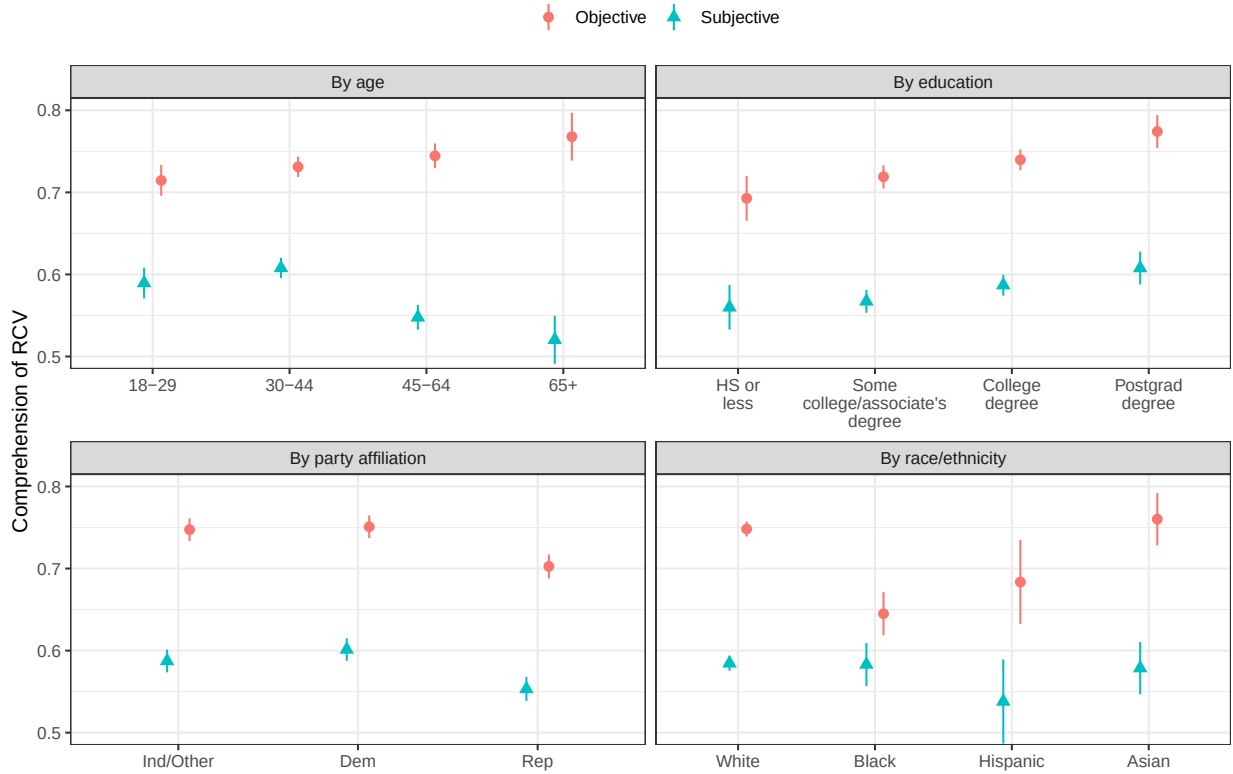
Although not the focus of this paper, we also report unconditional differences in mean comprehension by education, party, and race in Figure 5 (with 95% confidence intervals). The figures confirm that both objective and subjective comprehension are increasing in education. Unconditional on covariates, Republicans have lower average self-reported and objective understanding of RCV, which could help explain (or arise from) the previously noted partisan difference in opinion toward RCV. Racial gaps in objective understanding

Table 1: RCV comprehension (self-reported and objective) as a function of information treatments and covariates (OLS)

	DV: RCV comprehension [0-1]			
	Self-reported comprehension		Objective comprehension	
	(1)	(2)	(3)	(4)
Intercept	0.570*** (0.009)	0.559*** (0.022)	0.675*** (0.007)	0.606*** (0.018)
Treatment: Infographic	-0.030* (0.012)	-0.028* (0.012)	0.065*** (0.010)	0.065*** (0.010)
Treatment: Video	0.063*** (0.012)	0.062*** (0.012)	0.111*** (0.010)	0.109*** (0.010)
Some college/associate's deg.		0.006 (0.018)		0.014 (0.015)
College degree		0.011 (0.018)		0.033* (0.015)
Postgraduate degree		0.023 (0.020)		0.051** (0.016)
Democrat		0.007 (0.011)		-0.003 (0.009)
Republican		-0.016 (0.012)		-0.032** (0.010)
Heard of RCV		0.099*** (0.010)		0.113*** (0.008)
Voted in RCV election		0.019 (0.017)		-0.094*** (0.019)
Age: 30-44		0.009 (0.013)		0.009 (0.011)
Age: 45-64		-0.035* (0.014)		0.028* (0.012)
Age: 65+		-0.068** (0.021)		0.041* (0.018)
Woman		-0.074*** (0.010)		-0.020* (0.008)
Num.Obs.	3020	3016	3020	3016
R2	0.021	0.102	0.039	0.122
R2 Adj.	0.020	0.099	0.038	0.119

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Figure 5: Unconditional differences in comprehension of RCV



of RCV are more striking than gaps in self-reported understanding of RCV (notably for White and Black respondents), suggesting that previous research based on self-reported understanding (e.g. [Donovan et al. 2019](#)) may have understated racial gaps in comprehension.

4.2 Attitudes toward RCV vs plurality

Figure 6 summarizes our respondents' relative evaluative judgments of RCV and plurality, separately by treatment group. Each point shows the proportion of respondents who indicated the response on the horizontal axis (prefer the plurality outcome, no preference, or prefer the RCV outcome) separately by treatment group (a different color and point character for each group), aspect being evaluated (fairness of outcome in the top row and winner's mandate to govern on the bottom), and scenario (partisan on the left and non-partisan on the right). The top left panel shows that, when asked which outcome would be more fair in the partisan scenario, around 60% of respondents in the control group favored

the plurality outcome, around 30% favored the RCV outcome, and 10% favored neither (i.e. they responded “Don’t know/prefer not to answer”). By contrast, the corresponding numbers for the video group were 49%, 42%, and 9%, indicating that the video treatment caused respondents to be much more likely to view the RCV outcome as more fair. The infographic treatment appears to have had a similar, though slightly smaller, effect. The pattern for fairness is similar in the non-partisan scenario and on the mandate questions, though a substantially larger proportion of respondents chose not to express a preference between the RCV outcome and plurality outcome on the mandate questions. This could indicate that respondents are unsure how to interpret the concept of “mandate to govern” and/or that respondents interpret the concept as related to how *other* voters judge the election result and are unsure what to think.

Figure 6: Distribution of evaluative judgments by question and information treatment

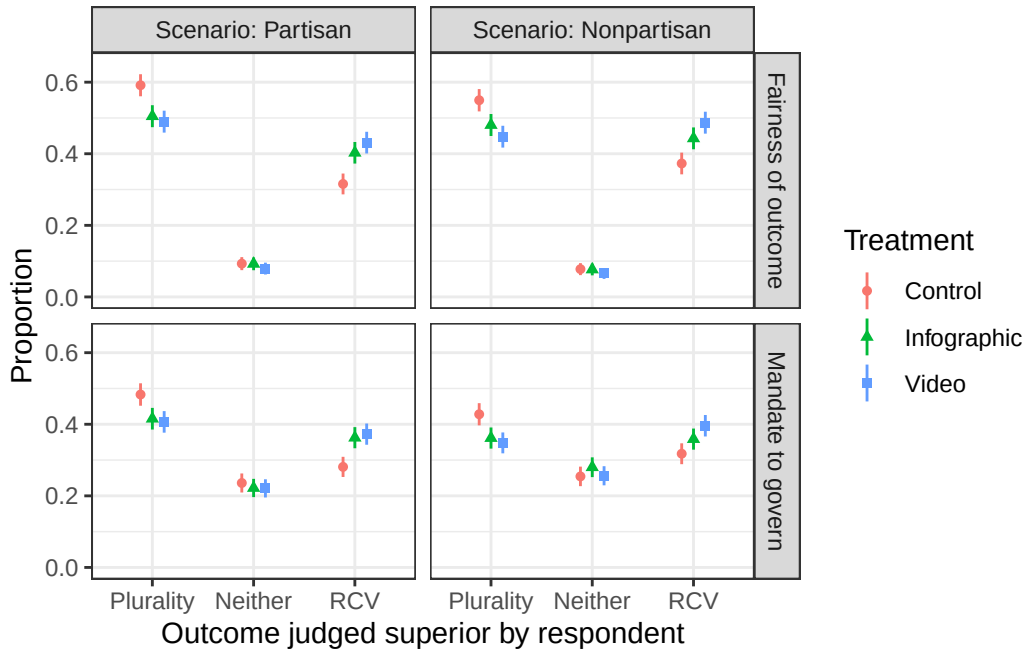


Table 2 reports regression results measuring the effect of our information treatments on the perceived relative fairness of the RCV and plurality outcome in our hypothetical election scenarios. Recall that our fairness index is a score between -0.5 and 0.5 that represents the degree to which respondents view the RCV outcome as more fair than the plurality outcome, averaging the two scenarios. Table 2 shows the results of regressing

Table 2: Effect of information treatments on perception that RCV outcome (vs plurality outcome) is more fair (OLS)

	DV: Perceived fairness of RCV outcome vs plurality outcome [-.5, .5]					
	Sample: All		Sample: Dems		Sample: Reps	
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment: Infographic	0.078*** (0.019)	0.081*** (0.018)	0.112*** (0.032)	0.108*** (0.029)	0.069* (0.029)	0.069* (0.029)
Treatment: Video	0.108*** (0.019)	0.111*** (0.018)	0.143*** (0.032)	0.133*** (0.030)	0.084** (0.030)	0.084** (0.029)
Covariates	No	Yes	No	Yes	No	Yes
Num.Obs.	3020	3016	1095	1095	986	984
R2	0.011	0.157	0.020	0.188	0.009	0.042
R2 Adj.	0.010	0.153	0.018	0.180	0.007	0.032

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

this fairness index on treatment status, with and without covariates.¹⁰ In analysis with all respondents (columns 1 and 2), we estimate that the video treatment increases the fairness index compared to control by about .085, as would be observed if the video caused 8.5% of respondents to move from favoring the plurality outcome to favoring the RCV outcome. The estimated effect for the infographic is smaller at about .06. We note that video also had a larger effect on comprehension than the infographic did; together these findings suggest that the treatments affected perceived fairness of RCV outcomes by increasing comprehension of the RCV procedure.

Following our PAP, we also report results separately by respondent party (columns 3-6 in Table 2). The results suggest that the effect of our treatments is larger for Democrats than for Republicans. This difference could be due to Republicans' higher average skepticism of RCV (shown below in Figure 7), or it could be due to differences between the groups on other dimensions (e.g. education).

Table 3 reports results where the outcome is our mandate index. The effect sizes are slightly smaller than for the fairness index, but the qualitative results are the same: larger effects for the video treatment than for the infographic treatment; larger effects for

¹⁰As above, the covariates include gender, age group, education level, political party affiliation, and whether the respondent had previously heard of or voted in an RCV election. Full regression results appear in Appendix Table 5; weighted results appear in Table 13.

Table 3: Effect of information treatments on perception that RCV winner (vs plurality winner) has stronger mandate to govern (OLS)

	DV: Perceived mandate to govern for RCV winner vs plurality winner [-.5, .5]					
	Sample: All		Sample: Dems		Sample: Reps	
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment: Infographic	0.064*** (0.017)	0.067*** (0.016)	0.077** (0.028)	0.075** (0.027)	0.059* (0.028)	0.061* (0.028)
Treatment: Video	0.082*** (0.017)	0.084*** (0.016)	0.102*** (0.028)	0.095*** (0.027)	0.067* (0.029)	0.071* (0.029)
Covariates	No	Yes	No	Yes	No	Yes
Num.Obs.	3020	3016	1095	1095	986	984
R2	0.008	0.078	0.013	0.113	0.006	0.029
R2 Adj.	0.008	0.074	0.011	0.104	0.004	0.018

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Democrats than for Republicans.

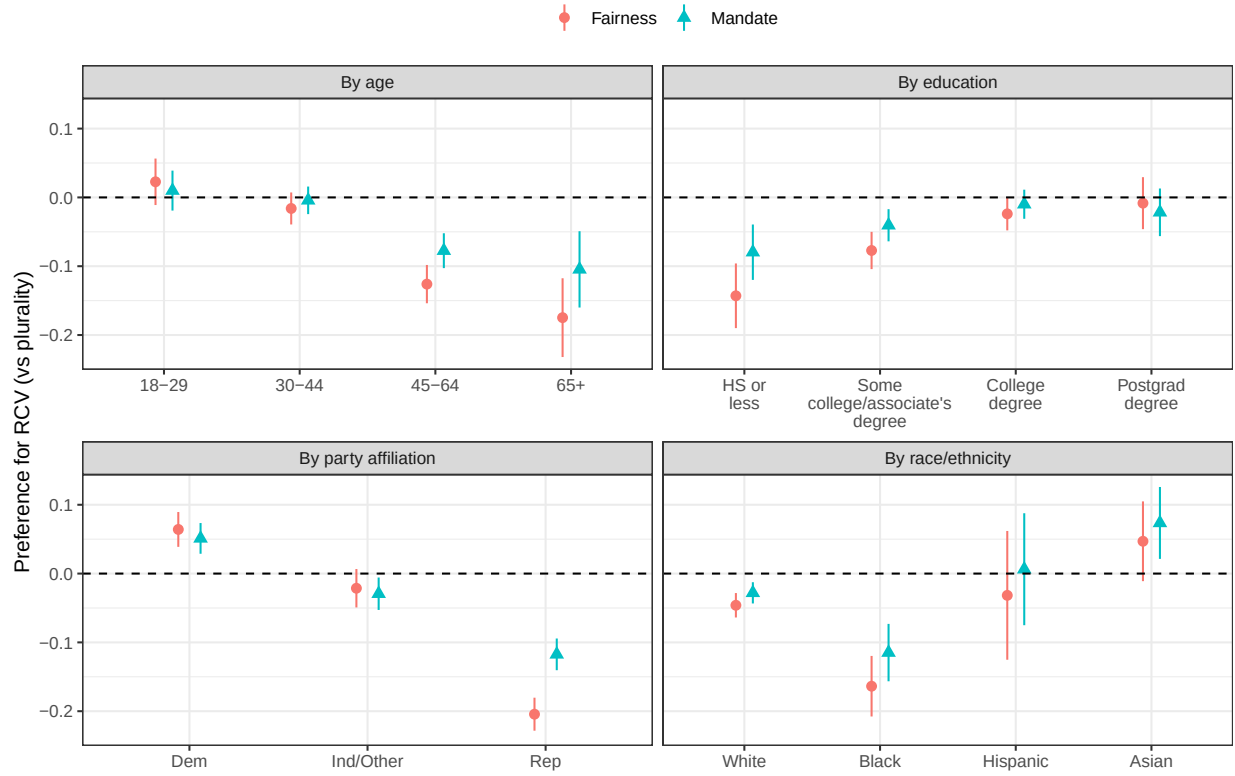
Figure 7 shows the fairness and mandate indices (unconditional on covariates) for some relevant categories of respondents. The figures confirm that favorability toward RCV is higher for younger people, more educated people, and more liberal people; it is also lower for Black people than for other groups.

4.3 Effect of comprehension on attitudes

We now turn to the question of whether individuals who better understand RCV are more inclined to view RCV election outcomes as fair (relative to the corresponding plurality outcomes). We estimate this relationship in two ways. First, we run ordinary least squares (OLS) regressions (“selection on observables”) of our perceived-fairness index on respondents’ measured comprehension of RCV, with and without key covariates. Second, we run an instrumental-variables (IV) analysis in which receiving either the infographic or the video treatment serve as a binary instrument for RCV comprehension.

Columns 1 and 2 of Table 4 report the OLS estimates of perceived fairness on our RCV comprehension index. Column 1, which does not include covariates, indicates that answering one additional question correctly (an increase of .25 in the comprehension index)

Figure 7: Unconditional differences in perceived fairness/mandate of RCV outcome relative to plurality outcome



is associated with a .12-point increase in the perceived fairness index. Adding covariates (namely education, political party affiliation, prior exposure to RCV, age, and gender) reduces this coefficient by about 25%, suggesting that the type of respondent who tends to score well in comprehension also tends to respect RCV outcomes. (Full regression results appear in Appendix Table 6, and with population weights in Appendix Table 14.)

Although these OLS regressions suggest that understanding RCV better causes respondents to view RCV outcomes more favorably, these estimates may be biased by unmeasured confounding. It could be the case that respondents who are predisposed to support RCV may both learn more about the system and be more inclined to view its outcomes as fair

To address this possible confounding, we ran the IV analysis described in our pre-analysis plan. We use a binary indicator for receiving either the infographic or video treatment (as opposed to the control text alone) as an instrument for respondents' comprehension of RCV, proxied by their comprehension index. The exclusion restriction in this case requires

Table 4: Effect of RCV comprehension score on perception that RCV outcome is fair (vs plurality outcome is fair): OLS and IV

	DV: Perceived fairness of RCV outcome vs plurality outcome [-.5, .5]			
	OLS		IV	
	(1)	(2)	(3)	(4)
RCV comprehension score	0.493*** (0.031)	0.372*** (0.031)	1.062*** (0.193)	1.104*** (0.189)
Covariates	No	Yes	No	Yes
Num.Obs.	3020	3016	3020	3016
R2	0.069	0.181	-0.023	0.043
R2 Adj.	0.068	0.178	-0.023	0.039

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

that being assigned to treatment affects perceived fairness of the RCV outcome only by affecting understanding of RCV. The exclusion restriction could be violated, for example, if the treatments conveyed positive messages about RCV in addition to explanation of the RCV mechanism. We chose treatments we thought were tonally neutral and solely focused on the RCV algorithm.¹¹

Columns 3 and 4 of Table 4 present the two-stage least squares (2SLS) results.¹² In the specification without covariates (Column 3), the coefficient on “RCV comprehension score” is 1.062 (SE = 0.193, p < 0.001), roughly double the corresponding OLS estimate. When we add the same set of covariates as in the OLS (Column 4),¹³ shows the estimated effect remains highly significant and similar in magnitude, at 1.104 (SE = 0.189, p < 0.001).¹⁴ We note that the IV estimates are over twice as large as their OLS counterparts, despite the fact that including covariates reduced the OLS estimates (which suggests that remaining selection bias in the OLS is probably positive). It is possible that there is an exclusion restriction violation (e.g. because the treatments made respondents feel more open to alternative election methods, or convinced them that the researchers want them to answer

¹¹To interpret the IV estimate as a local average treatment effect, we also need a monotonicity assumption, which in this case states that no one’s understanding of RCV is reduced by the treatment.

¹²The instrument is strong, with an F-statistic of 100.6.

¹³Table 7 shows the full 2SLS results including coefficients on covariates.

¹⁴Note that the reported R^2 for the IV specifications can be negative or quite small when computed in the standard OLS manner, which is not unusual in 2SLS and does not indicate a problem per se.

favorably to RCV) despite our efforts to design the treatments to be factually focused and tonally neutral. The LATE estimand may also be larger than the ATE, e.g. because it focuses on the effect of comprehension on perceived fairness for those who learn the most from the treatment; these respondents may be more receptive to changing their mind about RCV compared to those whose comprehension is harder to manipulate experimentally. We suspect an important reason for the difference is that our RCV comprehension score is a noisy measure of understanding, so that our OLS estimates are biased downwards and our IV estimates more closely approximate the effect of greater understanding.¹⁵

Together, both the OLS and the IV analyses are consistent with the idea that better comprehension of the mechanics of RCV would make Americans more likely to regard RCV outcomes as fair. Although questions remain about the magnitude of the effect (given that violation of the exclusion restriction cannot be ruled out) the overall pattern of results suggests that comprehension of RCV rules is an important condition for viewing RCV outcomes as legitimate.

5 Discussion

This study assessed whether enhancing voters’ comprehension of ranked choice voting could increase public support for RCV outcomes, particularly in scenarios (emphasized by advocates) where RCV and plurality voting produce might produce different winners. While previous research has highlighted a correlation between self-reported understanding of RCV and satisfaction with RCV, the causality of this relationship remained in doubt; moreover, previous work had not measured objective understanding of RCV or assessed its role in RCV acceptance. Our experimental design addressed these issues, producing a clear conclusion: better objective comprehension of RCV appears to meaningfully improve voter attitudes toward RCV outcomes.

¹⁵If the endogenous treatment is “understanding of RCV”, then this is a problem of measurement error in the OLS regression. If the endogenous treatment is “proportion of comprehension questions correctly answered”, then this is a violation of the exclusion restriction.

Its results demonstrate that both a video and an explanatory infographic produced notable increases in objective understanding of RCV, with voters in the video group showing the largest improvement. Using OLS and IV strategies we also found evidence that greater objective comprehension produces more favorable evaluations of RCV outcomes

We showed that both a video and an explanatory infographic produced notable increases in objective understanding of RCV and in perceived fairness of the RCV outcome in scenarios where it conflicts with plurality; for both outcomes (comprehension and perceived fairness), voters in the video group show the largest improvement. Using OLS we found that greater objective comprehension is associated with more favorable evaluations of RCV outcomes, including when we control for age, gender, education, party affiliation, and prior exposure to RCV. Moreover, leveraging treatment assignment as an instrument for RCV comprehension, we obtain further evidence that greater comprehension produces substantially higher support for RCV outcomes.

Ultimately, our results highlight a key insight into electoral system reform: public attitudes toward institutional change are shaped not only by partisanship or self-interest, but also by the quality of information and voter understanding. As we have shown, implementing low-cost tools like videos and infographics can make sense of how ranked choice elections work and may, in turn, increase support for outcomes that better reflect majority preferences

We also provide new insights into how different modes of information affect learning. The video treatment led to the largest gains in both comprehension and perceived fairness, while the infographic produced modest improvements in objective comprehension but paradoxically lowered self-reported understanding. One interpretation is that static visuals helped expose gaps in respondents' prior beliefs without fully resolving them, thereby undermining confidence. This disconnect between objective and subjective understanding echoes findings in cognitive psychology on "fluency illusions" and reinforces the importance of using behavioral measures (like our objective comprehension measure) in studies of political learning. Our results also suggest that videos may be more effective than text or

infographics at improving understanding and shaping attitudes

Our findings highlight the importance of leveraging educational materials in improving public understanding of voting systems. Although the proliferation of explainer videos on RCV and other subjects suggests that many journalists and reformers already believe that such videos can shift understanding about RCV and possibly build support for it, our findings provide the first clear evidence on these effects. Our results also provide important context that highlights the limitations of educational campaigns. For example, the full OLS regression results in Column 2 of Table 6 show a difference in the RCV fairness index between Democrats and Republicans (conditional on comprehension and covariates) of .2 out of a possible 1 (about .5 sds); the same regression implies that erasing this gap would require increasing Republicans’ RCV comprehension scores relative to Democrats’ by over .5 points out of a possible 1 (2 sds). Moreover, we find that Republicans’ assessments of RCV are less influenced by our information treatments than Democrats’ are. This underscores the reality that, although voters’ perception depend on understanding (which we have shown to be malleable), more engrained factors such as partisanship play a large role.

There are several important limitations to our research, some of which can be addressed by future work. First, although we show evidence that a video was more effective than an infographic, the two presentations differed in several ways including how they explained RCV, the amount of content, and aesthetics,¹⁶ so we cannot confidently isolate the role of the delivery format. Future research could more systematically compare the effectiveness of alternative ways of communicating about electoral systems to different audiences, including using different media and types of explanation and testing different types of messengers, to determine what makes voter education most effective. Second, our study (like [Cerrone & McClintock \(2023\)](#)) focuses on participants’ perceptions of hypothetical election results; in contrast, several other studies ask survey respondents to fill out a ballot in a hypothetical election to mimic the voter experience (e.g. [Blais et al. 2021](#), [Anthony et al. 2024](#)). While

¹⁶We also disabled the “Next page” button for 30 seconds for the video treatment group to encourage watching the video, but we did not apply a similar restriction for the infographics group.

these studies suggest that the mere act of ranking candidates has little appeal on average (at least for US voters), future studies could build on ours to ask whether, with a better understanding of how RCV works, voters would feel more positively toward ranking candidates. Third, our study does not aim to assess the US public’s acceptance of RCV in general; instead it focuses on acceptance of RCV vs plurality outcomes in the important situation where the two might conflict. Future studies could study how understanding affects this broader acceptance, perhaps using behavioral survey-based measures such as willingness to sign a petition or even turnout or valid voting in real elections. Fourth, we find substantial immediate effects of better RCV comprehension on support for RCV outcomes. Future work could assess whether these effects persist or spill over onto support for other voting systems. Fifth, our online sample skews younger and more educated than the US voting population; even adjusting for observable factors via weighting, it is likely more politically engaged and savvy with video and text than the target population. This could exaggerate both baseline support for RCV and the effect of our treatments.

Despite these limitations, the study offers several contributions. First, it shows that short explanatory materials can improve objective comprehension of RCV. Second, it demonstrates that better understanding of RCV’s mechanics produces increased support for RCV’s outcomes in cases where those outcomes have been most controversial, i.e. where vote transfers reverse the initial ordering of candidates. Third, it highlights that self-reported comprehension may diverge substantially from actual comprehension, and that relying on the former (as past studies have done) can understate gaps in knowledge and mischaracterize the effectiveness of education efforts.

Our results also have broader implications for understanding how citizens form judgments about electoral fairness. When voters misunderstand how an election rule works, they may reject its outcomes even when those outcomes reflect majority preferences. Improving public understanding of electoral reforms is therefore critical not just for informed participation in new systems but for the legitimacy of the electoral process itself.

6 References

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7 Appendix

7.1 Details on video treatment

Link: <https://www.youtube.com/watch?v=oHRPMJmzBBw>

Produced by: Molly Bloom and Curtis Gilbert Photographed by: Jeffrey Thompson

Transcript

Pick your favorite color – the ranked choice voting way!

Instead of voting for just one color you get to rank your top three.

(Voter: “Well, purple is the best, but if I can’t have purple I want blue, and if neither of those wins I guess I can live with orange.”)

Now, let’s count up everybody’s votes. Under ranked choice voting rules, it’s not enough just to get the most votes. You need a majority – more than 50% of the votes.

Purple’s ahead but it has only seven votes. It needs at least 11 to win. So we eliminate the color in last place (sorry orange fans! we’re going to your second choice). Two more for green, one for purple. But no color has 11 votes yet – still no majority. Bye bye blue! One more for purple, four for green. And we have a winner – the ranked choice voting way.

7.2 Full regression tables

Table 5: Effect of information treatments on perceived fairness of RCV outcome vs plurality outcome, overall and by party affiliation (OLS) - full regression results

	DV: Perceived fairness of RCV outcome vs plurality outcome [-.5, .5]					
	Sample: All		Sample: Dems		Sample: Reps	
	(1)	(2)	(3)	(4)	(5)	(6)
Intercept	-0.113*** (0.014)	-0.161*** (0.032)	-0.023 (0.024)	-0.157** (0.056)	-0.256*** (0.020)	-0.199*** (0.049)
Treatment: Infographic	0.078*** (0.019)	0.081*** (0.018)	0.112*** (0.032)	0.108*** (0.029)	0.069* (0.029)	0.069* (0.029)
Treatment: Video	0.108*** (0.019)	0.111*** (0.018)	0.143*** (0.032)	0.133*** (0.030)	0.084** (0.030)	0.084** (0.029)
Some college/associate's deg.		0.052+ (0.027)		-0.005 (0.051)		0.045 (0.040)
College degree		0.077** (0.026)		0.063 (0.049)		0.007 (0.039)
Postgraduate degree		0.079** (0.030)		0.036 (0.054)		0.054 (0.047)
Democrat		0.060** (0.018)				
Republican		-0.146*** (0.019)				
Heard of RCV		0.208*** (0.016)		0.335*** (0.026)		0.065* (0.027)
Voted in RCV election		-0.062* (0.030)		-0.100* (0.043)		0.010 (0.055)
Age: 30-44		-0.049* (0.020)		-0.046 (0.029)		-0.078* (0.038)
Age: 45-64		-0.129*** (0.021)		-0.103** (0.035)		-0.134*** (0.038)
Age: 65+		-0.186*** (0.032)		-0.200*** (0.059)		-0.196*** (0.048)
Woman		-0.033* (0.015)		-0.044+ (0.024)		-0.029 (0.025)
Num.Obs.	3020	3016	1095	1095	986	984
R2	0.011	0.157	0.020	0.188	0.009	0.042
R2 Adj.	0.010	0.153	0.018	0.180	0.007	0.032

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 6: Effect of objective RCV comprehension on perceived fairness of RCV outcome vs plurality outcome, overall and by party affiliation (OLS) - full regression results

	DV: Perceived fairness of RCV outcome vs plurality outcome [-.5, .5]			
	Unweighted		Weighted	
	(1)	(2)	(3)	(4)
Intercept	-0.412*** (0.023)	-0.341*** (0.036)	-0.398*** (0.037)	-0.310*** (0.053)
RCV comprehension score	0.493*** (0.031)	0.372*** (0.031)	0.403*** (0.051)	0.318*** (0.053)
Some college/associate's deg.		0.048+ (0.026)		0.060+ (0.035)
College degree		0.063* (0.025)		0.070* (0.035)
Postgraduate degree		0.060* (0.029)		0.035 (0.042)
Democrat		0.063*** (0.018)		0.084** (0.030)
Republican		-0.133*** (0.018)		-0.110*** (0.030)
Heard of RCV		0.166*** (0.016)		0.130*** (0.027)
Voted in RCV election		-0.029 (0.029)		-0.022 (0.051)
Age: 30-44		-0.055** (0.019)		-0.042 (0.033)
Age: 45-64		-0.139*** (0.021)		-0.120*** (0.034)
Age: 65+		-0.201*** (0.033)		-0.170*** (0.044)
Woman		-0.028+ (0.015)		-0.039 (0.025)
Num.Obs.	3020	3016	3020	3016
R2	0.069	0.181	0.046	0.147
R2 Adj.	0.068	0.178	0.046	0.143

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 7: Effect of objective RCV comprehension on perceived fairness of RCV outcome vs plurality outcome (IV) - full regression results

	DV: Perceived fairness of RCV outcome vs plurality outcome [-.5, .5]			
	Unweighted		Weighted	
	(1)	(2)	(3)	(4)
Intercept	-0.830*** (0.142)	-0.830*** (0.130)	-0.788*** (0.143)	-0.722*** (0.139)
RCV comprehension score	1.062*** (0.193)	1.104*** (0.189)	0.934*** (0.194)	0.936*** (0.204)
Some college/associate's deg.		0.037 (0.029)		0.043* (0.022)
College degree		0.041 (0.028)		0.040+ (0.024)
Postgraduate degree		0.024 (0.033)		-0.006 (0.028)
Democrat		0.063*** (0.019)		0.092*** (0.019)
Republican		-0.111*** (0.020)		-0.087*** (0.021)
Heard of RCV		0.083** (0.027)		0.074** (0.025)
Voted in RCV election		0.043 (0.036)		0.030 (0.038)
Age: 30-44		-0.059** (0.021)		-0.044 (0.027)
Age: 45-64		-0.160*** (0.023)		-0.147*** (0.027)
Age: 65+		-0.232*** (0.036)		-0.192*** (0.028)
Woman		-0.011 (0.016)		-0.024 (0.016)
Num.Obs.	3020	3016	3020	3016
R2	-0.023	0.043	-0.034	0.046
R2 Adj.	-0.023	0.039	-0.035	0.043

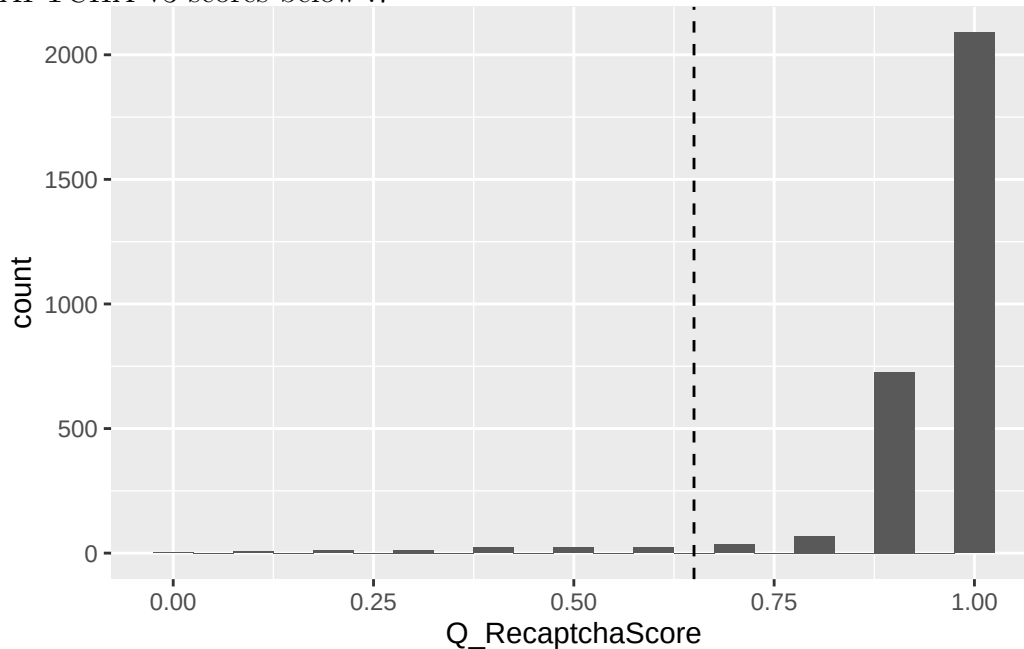
+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

7.3 Results without possible bots

Our primary analysis does not exclude any respondents. The purpose of including all respondents is to ensure that the control group and treatment groups are fully randomized: removing a particular type of respondent based on post-treatment behavior can make treatment groups less comparable and introduce bias.

As stated in our PAP, we reproduce in the appendix our primary analysis (effect of treatment on comprehension, effect of treatment on perceived fairness of RCV outcomes) excluding respondents with low reCAPTCHA v3 scores (below .7). reCAPTCHA v3 is a bot detection API provided by google that determines whether a website visitor has “suspicious” (i.e. bot-like) actions judging by the pattern of mouse and keyboard inputs.

This figure shows the distribution of reCAPTCHA scores. 136 respondents have reCAPTCHA v3 scores below .7



As stated in the PAP, we also present regressions of reCAPTCHA v3 scores (reverse-coded) and a binary exclusion variable (score $< .7$) on treatment status. If these outcomes are related to treatment status, it suggests a problem with bot detection, and also a reason to be concerned about imbalance across treatment groups in the restricted sample.

The p-value on an F-test for joint significance of the treatment predictors in the first

Table 8: Effect of information treatments on (reverse-coded) reCAPTCHA v3 scores

	DV: bot-liess score	DV: excluded due to bot-liess
Intercept	0.055*** (0.004)	0.030*** (0.005)
Treatment: infographic	-0.003 (0.005)	-0.008 (0.007)
Treatment: video	-0.006 (0.006)	-0.002 (0.007)
Num.Obs.	3020	3020
R2	0.000	0.000
R2 Adj.	0.000	0.000

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

regression (DV: reverse-coded reCaptcha v3 score, i.e. “bot-liess score”) is 0.451. The p-value on an F-test for joint significance of the treatment predictors in the second regression (DV: reCaptcha v3 score below .7, i.e. “bot-liess score < .7”) is 0.512.

We stated in our PAP that if we detected that treatment affected exclusion we would investigate imbalance across treatment groups by regressing treatment status on covariates (“omnibus placebo test”). As it stands, do not find evidence that reCAPTCHA v3 scores are affected by treatment status, so we report our main results on the restricted sample.

Qualitatively our results closely match those with no exclusions.

Table 9: Effect of information treatments on comprehension of RCV (self-reported and objective, OLS) - possible bots excluded

	DV: RCV comprehension [0-1]			
	Self-reported comprehension		Objective comprehension	
	(1)	(2)	(3)	(4)
Intercept	0.570*** (0.009)	0.560*** (0.023)	0.680*** (0.007)	0.605*** (0.018)
Treatment: Infographic	-0.030* (0.012)	-0.028* (0.012)	0.062*** (0.010)	0.063*** (0.010)
Treatment: Video	0.063*** (0.012)	0.061*** (0.012)	0.108*** (0.010)	0.106*** (0.010)
Some college/associate's deg.		0.005 (0.018)		0.017 (0.015)
College degree		0.010 (0.018)		0.039** (0.015)
Postgraduate degree		0.019 (0.020)		0.059*** (0.017)
Democrat		0.010 (0.012)		0.001 (0.010)
Republican		-0.015 (0.012)		-0.023* (0.010)
Heard of RCV		0.102*** (0.010)		0.116*** (0.008)
Voted in RCV election		0.014 (0.018)		-0.086*** (0.021)
Age: 30-44		0.012 (0.013)		0.007 (0.011)
Age: 45-64		-0.035* (0.014)		0.025* (0.012)
Age: 65+		-0.071*** (0.021)		0.040* (0.018)
Woman		-0.074*** (0.010)		-0.022** (0.008)
Num.Obs.	2919	2915	2919	2915
R2	0.021	0.105	0.037	0.126
R2 Adj.	0.020	0.101	0.037	0.122

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 10: Effect of information treatments on perceived fairness of RCV, overall and by party affiliation (OLS) - possible bots excluded

	DV: Perceived fairness of RCV outcome vs plurality outcome [-.5, .5]					
	Sample: All		Sample: Dems		Sample: Reps	
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment: Infographic	0.082*** (0.019)	0.083*** (0.018)	0.113*** (0.033)	0.110*** (0.029)	0.067* (0.030)	0.065* (0.030)
Treatment: Video	0.113*** (0.019)	0.115*** (0.018)	0.146*** (0.033)	0.135*** (0.030)	0.090** (0.030)	0.089** (0.030)
Covariates	No	Yes	No	Yes	No	Yes
Num.Obs.	2919	2915	1055	1055	942	940
R2	0.012	0.163	0.021	0.193	0.010	0.046
R2 Adj.	0.011	0.159	0.019	0.185	0.008	0.035

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 11: Effect of RCV comprehension on perceived fairness of RCV (OLS and IV) - possible bots excluded

	DV: Perceived fairness of RCV outcome vs plurality outcome [-.5, .5]			
	OLS		IV	
	(1)	(2)	(3)	(4)
RCV comprehension score	0.507*** (0.032)	0.380*** (0.032)	1.147*** (0.205)	1.178*** (0.202)
Covariates	No	Yes	No	Yes
Num.Obs.	2919	2915	2919	2915
R2	0.071	0.187	-0.042	0.026
R2 Adj.	0.071	0.183	-0.043	0.022

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

7.4 Results with population weights

Following our PAP, we generated population weights by raking to match the margins for the US electorate in the 2022 Cooperative Election Survey by income, education, sex, race, and party affiliation. To guard efficiency given possibly very high weights, we trim the weights at the 99th percentile.

Following the PAP, our paper focuses on the unweighted results, but we present weighted versions of the primary analysis in the Appendix.

Table 12: Effect of information treatments on comprehension of RCV (self-reported and objective, OLS) - with weights

	DV: RCV comprehension [0-1]			
	Self-reported comprehension		Objective comprehension	
	(1)	(2)	(3)	(4)
Intercept	0.567*** (0.015)	0.595*** (0.030)	0.673*** (0.012)	0.604*** (0.029)
Treatment: Infographic	-0.069*** (0.020)	-0.074*** (0.019)	0.069*** (0.017)	0.061*** (0.016)
Treatment: Video	0.018 (0.021)	0.014 (0.020)	0.107*** (0.017)	0.101*** (0.017)
Some college/associate's deg.		0.008 (0.025)		0.026 (0.020)
College degree		0.015 (0.024)		0.054** (0.020)
Postgraduate degree		0.015 (0.028)		0.070** (0.023)
Democrat		0.018 (0.020)		-0.012 (0.017)
Republican		-0.030 (0.020)		-0.033* (0.016)
Heard of RCV		0.076*** (0.016)		0.087*** (0.015)
Voted in RCV election		0.003 (0.028)		-0.073* (0.032)
Age: 30-44		0.016 (0.022)		0.007 (0.022)
Age: 45-64		-0.008 (0.023)		0.044* (0.022)
Age: 65+		-0.053+ (0.029)		0.036 (0.026)
Woman		-0.101*** (0.016)		-0.021 (0.014)
Num.Obs.	3020	3016	3020	3016
R2	0.019	0.106	0.037	0.113
R2 Adj.	0.019	0.102	0.036	0.109

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 13: Effect of information treatments on perceived fairness of RCV, overall and by party affiliation (OLS) - with weights

	DV: Perceived fairness of RCV outcome vs plurality outcome [-.5, .5]					
	Sample: All		Sample: Dems		Sample: Reps	
	(1)	(2)	(3)	(4)	(5)	(6)
Intercept	-0.160*** (0.022)	-0.158*** (0.045)	-0.055 (0.043)	-0.104 (0.084)	-0.297*** (0.028)	-0.228** (0.073)
Treatment: Infographic	0.048 (0.030)	0.042 (0.029)	0.072 (0.055)	0.074 (0.050)	0.084+ (0.047)	0.078+ (0.046)
Treatment: Video	0.115*** (0.031)	0.108*** (0.030)	0.115* (0.058)	0.089 (0.055)	0.073+ (0.041)	0.074+ (0.042)
Some college/associate's deg.		0.069+ (0.036)		-0.017 (0.069)		0.044 (0.056)
College degree		0.091* (0.035)		0.068 (0.064)		-0.009 (0.052)
Postgraduate degree		0.059 (0.042)		-0.001 (0.077)		0.011 (0.064)
Democrat		0.082** (0.031)				
Republican		-0.118*** (0.030)				
Heard of RCV		0.155*** (0.026)		0.282*** (0.047)		0.008 (0.042)
Voted in RCV election		-0.038 (0.055)		-0.072 (0.081)		0.032 (0.080)
Age: 30-44		-0.037 (0.035)		-0.066 (0.062)		0.006 (0.064)
Age: 45-64		-0.105** (0.035)		-0.072 (0.062)		-0.060 (0.065)
Age: 65+		-0.158*** (0.044)		-0.207** (0.076)		-0.100 (0.074)
Woman		-0.044+ (0.025)		-0.037 (0.042)		-0.056 (0.040)
Num.Obs.	3020	3016	1095	1095	986	984
R2	0.012	0.131	0.012	0.146	0.010	0.031
R2 Adj.	0.012	0.127	0.010	0.138	0.008	0.021

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Table 14: Effect of RCV comprehension on perceived fairness of RCV (OLS and IV) - with weights

	DV: Perceived fairness of RCV outcome vs plurality outcome [-.5, .5]			
	OLS		IV	
	(1)	(2)	(3)	(4)
RCV comprehension score	0.403*** (0.051)	0.318*** (0.053)	0.934*** (0.194)	0.936*** (0.204)
Covariates	No	Yes	No	Yes
Num.Obs.	3020	3016	3020	3016
R2	0.046	0.147	-0.034	0.046
R2 Adj.	0.046	0.143	-0.035	0.043

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

7.5 Additional results

Table 15: Effect of treatments on RCV comprehension - separately by party

	DV: Objective comprehension of RCV mechanics [0, 1]					
	Sample: All		Sample: Dems		Sample: Reps	
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment: Infographic	0.065*** (0.010)	0.065*** (0.010)	0.056** (0.017)	0.053** (0.016)	0.066*** (0.018)	0.072*** (0.018)
Treatment: Video	0.111*** (0.010)	0.109*** (0.010)	0.106*** (0.017)	0.099*** (0.016)	0.114*** (0.018)	0.114*** (0.018)
Covariates	No	Yes	No	Yes	No	Yes
Num.Obs.	3020	3016	1095	1095	986	984
R2	0.039	0.122	0.034	0.139	0.040	0.087
R2 Adj.	0.038	0.119	0.032	0.130	0.038	0.077

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001